Machine Learning in Time Series

Lecture 6: NLP, Attention, and Causal ML

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Agenda

1. Seminar Overview & Roadmap
2. Part 1: Text as Data (NLP in Macro)
3. Part 2: Causal Machine Learning
4. Lab Results: DML Local Projections
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Seminar Overview & Roadmap

The 6-Lecture Journey

Where we are

We have spent 5 lectures optimizing $\hat{y}$ (the forecast). Today we move from forecasting to frontier empirical workflows that combine high-dimensional prediction with credible inference.

Lecture 1: Historical Evolution, the Prediction Problem, and Shrinkage.

Lecture 2: Kernel Methods and Support Vector Machines (SVM).

Lecture 3: High-Dimensional Linear Dynamics (BVAR, Sparse VAR).

Lecture 4: Deep Learning for Sequences (RNN/LSTM).

Lecture 5: Trees, Boosting, and Ensembles.

Lecture 6: NLP, Attention, and Causal ML.

Part 1: Text as Data (NLP in Macro)

A Brief History of NLP in Econometrics

Before we dive into the math, it is crucial to understand how NLP evolved from basic linguistics to a core econometric tool.

The Four Eras of NLP:

- **1950s–1980s (Rule-Based):** Linguists wrote hard-coded grammar rules. Inflexible and unable to handle the nuance of real-world text.
- **1990s–2000s (Statistical / TF-IDF):** The shift to probability. Economists used word counts and frequency matrices. (e.g., Tetlock 2007 on news sentiment).
- **2013 (Dense Embeddings):** Word2Vec triggers a revolution. Words are no longer discrete counts; they are continuous vectors mapping semantic meaning.
- **2017–Present (Transformers & LLMs):** The "Attention is All You Need" paper allows models to process entire documents holistically, capturing deep context.

Glossary: TF-IDF

TF-IDF (Term Frequency-Inverse Document Frequency)

- **What it is:** A foundational statistical method used to evaluate how important a specific word is to a specific document within a larger corpus.
- **The Intuition:**
  - *Term Frequency (TF):* If "inflation" appears many times in a Fed speech, it is important to that speech.
  - *Inverse Document Frequency (IDF):* If "the" or "economy" appears in *every* Fed speech, it is not informative. IDF mathematically penalizes words that are too common.
- **The Math:** $W_{i,j} = \mathrm{tf}_{i,j} \times \log\left(\frac{N}{\mathrm{df}_i}\right)$

Appendix Glossary: Transformers & LLMs

Transformers

- **What it is:** The specific neural network architecture invented in 2017 ("Attention is All You Need").
- **The Breakthrough:** Unlike older models (RNNs/LSTMs) that read text sequentially (word-by-word) and suffer from "amnesia," Transformers process all words *simultaneously* using the **Self-Attention Mechanism**. This instantly captures long-range context.

LLMs (Large Language Models)

- **What they are:** Massive AI systems (GPT, Gemini) built on the Transformer architecture.
- **Why they matter:** Trained on vast datasets with billions of parameters, they develop deep implicit understandings of syntax and logic, serving as the ultimate feature extractors for unstructured data.

The NLP Pipeline: From Raw Text to Tensors

How do we actually transform a central bank speech into a vector X_t?

1. Tokenization: Text is split into discrete units (tokens) and mapped to integer IDs from a vocabulary $\mathcal{V}$. *"Inflation is high"* $\rightarrow$ `["Inflation", "is", "high"]` $\rightarrow$ `[341, 12, 892]`

2. Vectorization (Embedding Matrix): We define a mapping function E that takes an integer token $w_i \in \mathcal{V}$ and returns a dense continuous vector:

$$E : \mathcal{V} \rightarrow \mathbb{R}^d$$

where d is the embedding dimension (e.g., $d = 300$).

3. Aggregation (The Document Vector): To use this in a VAR model, we aggregate the word vectors into a single document vector X_t:

$$X_t = \frac{1}{N} \sum_{i=1}^{N} E(w_i)$$

Alternatives: TF-IDF weighting, topic models, sentence/document encoders.

The Rise of Unstructured Data

Historically, macroeconomists relied strictly on numerical aggregates (GDP, CPI, Unemployment). But market expectations are driven by *words*—specifically, Central Bank communication.

The NLP Objective

How do we map a 50-page Federal Open Market Committee (FOMC) press release into a numerical vector X_t that we can feed into a VAR or a Random Forest?

Deliverables: (i) a hawkishness index, (ii) topic proportions, (iii) a document embedding, each usable as X_t in forecasting/inference.

The Old Standard: Bag-of-Words Methods Count the frequency of specific terms to create a "Hawkishness Index" (*inflation, tightening* minus *growth, dovish*).

- *The Fatal Flaw:* Ignores context. "We will **not** combat **inflation**" is counted as hawkish simply because the word "inflation" appeared.

The Geometry of Language: Word Embeddings

To solve the context problem, modern NLP models language using high-dimensional geometry.

Word2Vec (Mikolov et al., 2013): Words are translated into dense numerical vectors (e.g., $\mathbb{R}^{300}$). Embeddings place words in a vector space where semantic and contextual similarity is reflected in Euclidean distance or Cosine Similarity.

The Math of Semantics: Because meaning is mapped to geometry, the vectors can be algebraically manipulated. The model learns relationships without being explicitly programmed:

$$V(\text{Lagarde}) - V(\text{ECB}) + V(\text{Fed}) \approx V(\text{Powell})$$

- **Caution:** Embeddings reflect the training corpus; economics-specific meaning can drift across regimes/time.

The Transformer: The Math of Attention

Word embeddings are static (the vector for "bank" is the same whether it's a river bank or a central bank). Large Language Models (like BERT or GPT) solve this using the **Self-Attention Mechanism**.

The Attention Equation:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

- **Query (Q):** What a specific word is "looking for" in the sentence.
- **Key (K):** What every other word "offers."
- **Value (V):** The actual semantic content of the word.

The Attention Intuition: The Filing Cabinet Analogy

How does $\text{softmax}(QK^T/\sqrt{d_k})V$ actually work in practice?

The Analogy: Imagine you are searching through a filing cabinet (the sentence).

- Your **Query** is the search term: *"I need the object of the verb 'raise'."*
- The **Keys** are the labels on the folders: *"Noun," "Subject," "Interest Rates."*
- You compute the dot product ($Q \cdot K^T$). High dot products mean a strong match. The Softmax function turns these matches into percentages (weights).
- You pull out the **Values** (the files inside) and multiply them by those percentages.

The Econometric Result: The network dynamically weighs the importance of every word relative to every other word, capturing context via data-driven interactions. It handles negation and long-range dependencies better than bag-of-words.

Text-as-Data Evaluation Pitfalls

Constructing X_t is only half the battle. How do we validate that our NLP vector is economically meaningful and statistically sound?

The Honest Evaluation Checklist:

1. **Temporal Leakage:** Ensure the text timestamp matches the exact information set. Be careful aligning statement release times vs. the narrow window of high-frequency market reactions.
2. **Stability & Drift:** Embeddings and topics can drift over time as language evolves. Perform a rolling refit of your model and show whether the constructed index (e.g., hawkishness) remains stable across regimes.
3. **Economic External Validity:** Does the measure correlate with reality? Correlate your text-derived index with known outcomes (e.g., surprise rate changes, yield curve moves, or high-frequency monetary shocks).

Part 2: Causal Machine Learning

The Mindset Shift: Prediction vs. Intervention

To understand modern Causal ML, we must first unlearn the standard time-series framework.

1. **Granger Causality (VAR):**
   - *Question:* Does knowing past FFR help predict future inflation?
   - *Reality:* It is purely predictive. December coat sales "Granger-cause" January snow. If you ban coat sales, it still snows. Granger does not isolate a manipulable structural mechanism.

2. **Structural Causality (Our Goal):**
   - *Question:* If the Fed forces an **unexpected** rate hike, how does inflation respond, holding all other macroeconomic history constant?

The Endogeneity Problem: Defining the Treatment (D)

In clinical trials, treatment is a binary pill. In our Local Projections lab, the treatment D_t is the **Federal Funds Rate (FFR)**. The target Y_{t+h} is the change in inflation.

The Challenge

The Fed does not set D_t randomly. They set it based on the macroeconomic history X_t (past inflation, unemployment).

If we simply regress future inflation on the raw FFR, we are capturing the Fed's reaction function, not the causal effect of the rate hike. We must isolate the **policy shock**—the variation in FFR that cannot be predicted by macro history.

The Classical Fix: Frisch-Waugh-Lovell (FWL) Theorem

To isolate this "unexpected" hike, classical econometrics relies on the FWL theorem to mathematically "partial out" the confounding history X_t.

The 3 Steps of FWL:

1. **Filter Y:** Regress Inflation (Y) on macro history (X). The residuals $\tilde{Y}$ represent "Unexplained Inflation."
2. **Filter D:** Regress the FFR (D) on macro history (X). The residuals $\tilde{D}$ represent a "policy shock" under the identification assumption.
3. **Isolate Causal θ:** Regress $\tilde{Y}$ on $\tilde{D}$.

Identification Warning

Interpreting $\tilde{D}_t$ as a policy shock requires an identification assumption: after conditioning on X_t, remaining variation in D_t is as-good-as-exogenous.

The Breakdown: Why Naive ML and Classical FWL Fail

Why can't we just run FWL with OLS, or look at the coefficients of a Lasso model?

1. **The Problem with Naive ML (Regularization Bias):**
   - ML models (Lasso, Random Forests) introduce shrinkage to optimize predictions. This aggressive bias ruins naive plug-in interpretations of causal parameters. Your $\hat{\beta}$ might be shrunk to zero artificially.

2. **The Problem with Classical FWL (The Linear Straitjacket):**
   - Classical FWL uses OLS to filter X_t. But the Fed does not react linearly. They react to regime changes and complex interactions. If OLS fails to model these non-linearities, the residual $\tilde{D}_t$ is contaminated with omitted variable bias.

The Modern Fix: Double Machine Learning (DML)

Double/Debiased Machine Learning (Chernozhukov et al., 2018) upgrades FWL for the 21st century by using high-capacity Machine Learning to execute the filtering steps.

The DML Innovations:

1. **Non-Linear Filtering (Nuisance Models):** We use Random Forests to estimate $E[Y|X]$ and $E[D|X]$. The ML autonomously maps all non-linearities and interactions in the macro history without us hand-coding them.
2. **Cross-Fitting (The Safeguard):** ML models overfit. If we overfit the Fed's reaction function, our shock $\tilde{D}_t$ becomes pure noise. We fix this by splitting the sample: train the RF on Sample A, and compute the residuals on Sample B.
3. **Orthogonal Score:** $\hat{\theta} = \left(\sum \tilde{D}_t^2\right)^{-1} \sum \tilde{D}_t \tilde{Y}_t$

DML in Time Series: Dependence & Overlap

Applying DML requires caution in time-series settings due to serial dependence.

Key Adjustments for Macro Data:

- **Block Cross-Fitting:** Do not split the sample randomly. Split the data by *contiguous time blocks* to preserve the dependence structure within folds and avoid future-to-past data leakage.
- **Standard Errors:** The orthogonal score allows standard inference, but you must account for serial correlation. Use HAC, block bootstrap, or cluster-by-time estimators depending on the setting.
- **The Overlap Check:** Always plot the distribution of $\tilde{D}_t$. If your nuisance model perfectly predicts the treatment (FFR), $\tilde{D}_t \approx 0$. You lack the variation ("overlap") needed to identify the causal effect.

Reading the Frontier: Consuming a DML Paper

As you read papers claiming causal effects using ML, use this checklist:

1. **What are D, Y, and X?** Clearly identify the treatment, the outcome, and the high-dimensional set of confounders.
2. **How do they justify identification?** DML is an estimation tool, *not* an identification strategy. The paper must still argue why D is exogenous conditional on X.
3. **What ML methods are used for nuisance?** Did they use Lasso, Random Forests, or Neural Nets? Did they tune the hyperparameters rigorously?
4. **How do they do cross-fitting + inference?** Check their fold-splitting strategy (especially if it is panel or time-series data) and whether they adjusted standard errors for clustering/autocorrelation.

Lab Results: DML Local Projections

The R Lab: DML Local Projections Workflow

Our R script executes this exact DML logic over horizons $h = 0, 6, 12, 18, 24$:

1. **Data & Targets:** Extract CPI and FFR. Target is the cumulative change in inflation $Y_{t+h} - Y_{t-1}$.
2. **Nuisance Models:** Train Random Forests (‘ranger‘ package) to predict both the target Y and treatment D using 36 lags of macro history.
3. **Cross-Fitting Function:** Split data in blocks. Train on Fold 1, predict out-of-sample residuals $(\tilde{Y}, \tilde{D})$ on Fold 2 to prevent overfitting bias.
4. **Causal Stage:** Run an intercept-free OLS (‘lm‘) of $\tilde{Y}$ on $\tilde{D}$ with robust standard errors (‘vcovHC‘). Use HAC / block-robust SE when dependence is material. The coefficient is the isolated structural impact.

Model Card: Structural Parameters (FFR $\rightarrow$ Inflation)

The table tracks the causal impact of a 1pp hike in FFR on the **cumulative change** in YoY Core CPI across various time horizons using DML.

Horizon (h)	Estimate (pp)	Std. Error	P-Value	Significant?
0 (Contemp.)	0.0036	0.0065	0.5859	No
6 Months	0.0571	0.0232	0.0142	**Yes**
12 Months	0.0795	0.0327	0.0152	**Yes**
18 Months	0.0384	0.0383	0.3166	No
24 Months	**-0.0160**	0.0416	0.7012	No

Analytical Insight: Persistence and Transmission

Short-horizon positive estimates are consistent with price puzzle patterns; interpretation depends on identification and measurement/timing of the policy shock. Crucially, at $h = 24$, the sign finally flips to negative, capturing the long-lagged structural transmission (demand destruction).

Decoding the IRF Figure

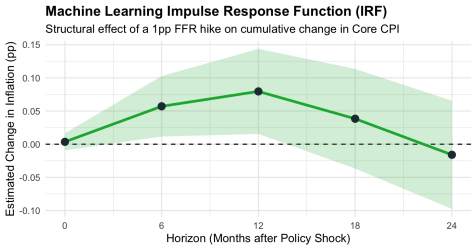

Figure 1: ML Impulse Response Function

How to read this plot:

- **X-Axis:** Months after the 1pp policy shock.
- **Y-Axis:** Estimated cumulative change in Core CPI (pp).
- **Green Line & Dots:** The $\hat{\theta}$ DML estimates.
- **Shaded Ribbon:** 95% Confidence Interval calculated via robust standard errors. When the ribbon entirely excludes the dashed zero line (e.g., months 6 and 12), the effect is statistically significant.

Seminar Conclusion

The Golden Rules of Time-Series ML

What have we learned about empirical workflows over the past 6 lectures?

- **Respect the Arrow of Time:** Never use random K-fold cross validation. Always use time-ordered rolling or expanding windows to prevent data leakage.
- **The Humility Check (AR1):** Always benchmark complex Ensembles or LSTMs against a simple, highly-persistent AR(1) model.
- **Stationarity Still Matters:** ML is not a magic wand for unit roots. Check unit roots/cointegration and model accordingly; differencing is one option but can discard long-run information.
- **Regularize Intelligently:** High capacity models will overfit macroeconomic noise. Use Shrinkage (Ridge/Lasso) or strict tree depth limits to force parsimony.

From Prediction to Policy

The Ultimate Lesson

Predictive power (minimizing RMSE, maximizing R^2) is fundamentally different from causal identification ($\hat{\theta}$).

The Modern Econometric Pipeline:

1. **Procure & Embed:** Gather numerical controls and unstructured (NLP) data.
2. **Predict (The $\hat{y}$ phase):** Use Ensembles/LSTMs to map complex state spaces out-of-sample.
3. **Identify (The $\hat{\theta}$ phase):** Use Double Machine Learning (DML) to partial out high-dimensional confounding via cross-fitting.
4. **Conclude:** Deliver $\sqrt{N}$-consistent structural parameters (like the ML-IRF) for robust policy decisions.

The Machine Learning Econometrician

The Paradigm Shift

Machine Learning does not replace econometric theory; it provides the computational muscle to execute it without restrictive parametric assumptions. We have moved from curve-fitting to structural identification.

Your Methodological Arsenal:

- **High-Dimensionality:** Shrinkage (Lasso/Ridge) to survive $P > N$.
- **Non-Linearity:** Trees, Ensembles, and Deep Learning to map complex state spaces.
- **Unstructured Data:** NLP and Attention to quantify human narratives.
- **Credible Inference:** DML to extract causal $\hat{\beta}$ from the predictive noise.

The frontier is yours.